

Introduction

We measured regional resilience “how areas withstand shocks” using Social Growth Index. Resilience scores are computed with the Copeland method. LISA maps reveal spatial dependence, and missing municipal GDP data is estimated using downscaling Fixed Rank Kriging.

$$\text{GDP} = \text{Population} \times \text{GDP Per Capita}$$



Research Questions

Evaluating the reciprocal influence of neighbouring provinces.

Measuring the resilience of country and regions of Europe and Italian provinces.

Determining the most resilient provinces.

Data



Layers	Where
Countries	Europe
Regions	Europe
Provinces	Italy
Municipality	Italy

Missing data

We used different data sources to obtain GDP and population figures. For the downscaling, we relied on the number of employees at the municipal level as a proxy for GDP.

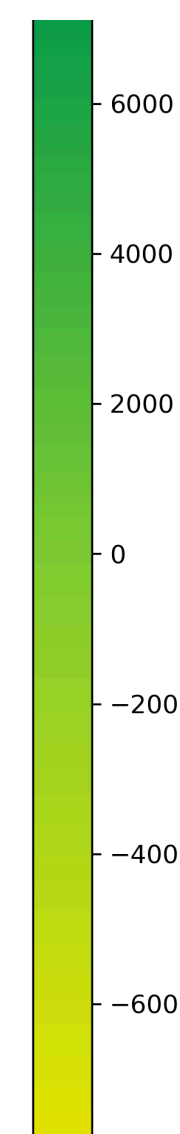
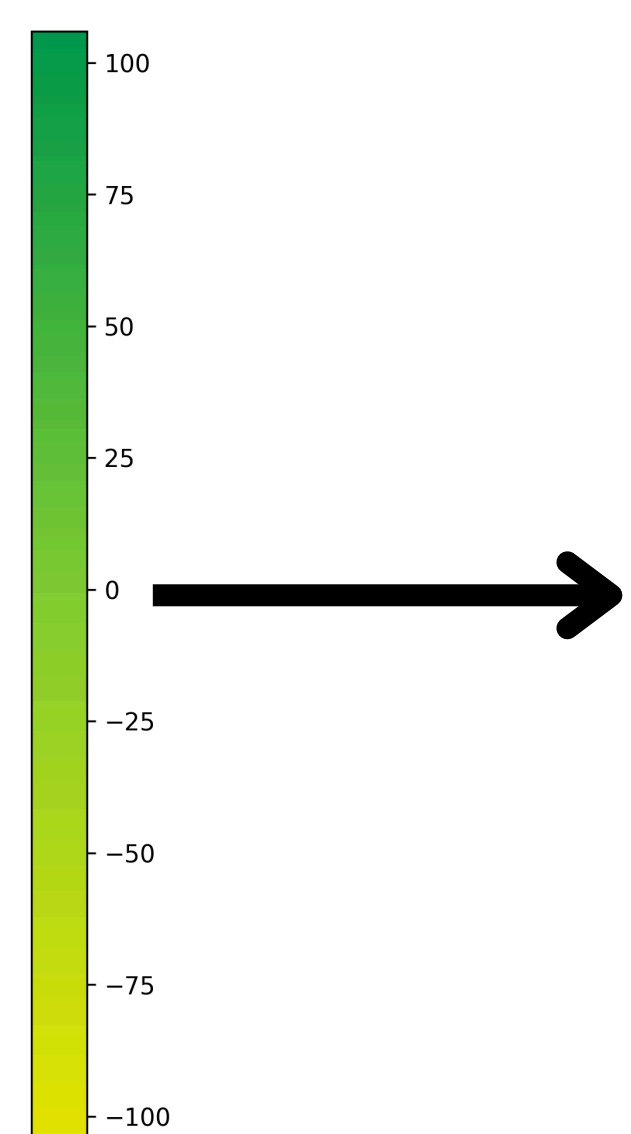
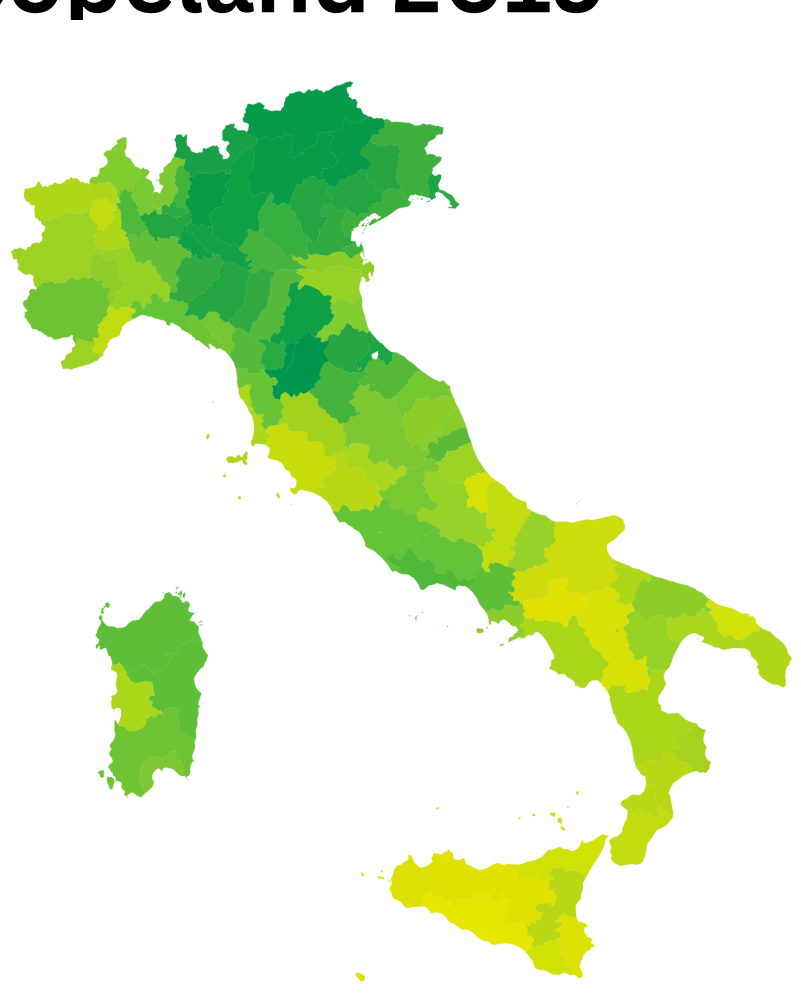
Copeland Method

Resilience is assessed using the Copeland method based on three indicators: GDP growth, GDP per capita growth, and population growth. A penalty term is applied to each indicator, computed as a weighted average between the territory’s value and the mean of its neighbouring regions.

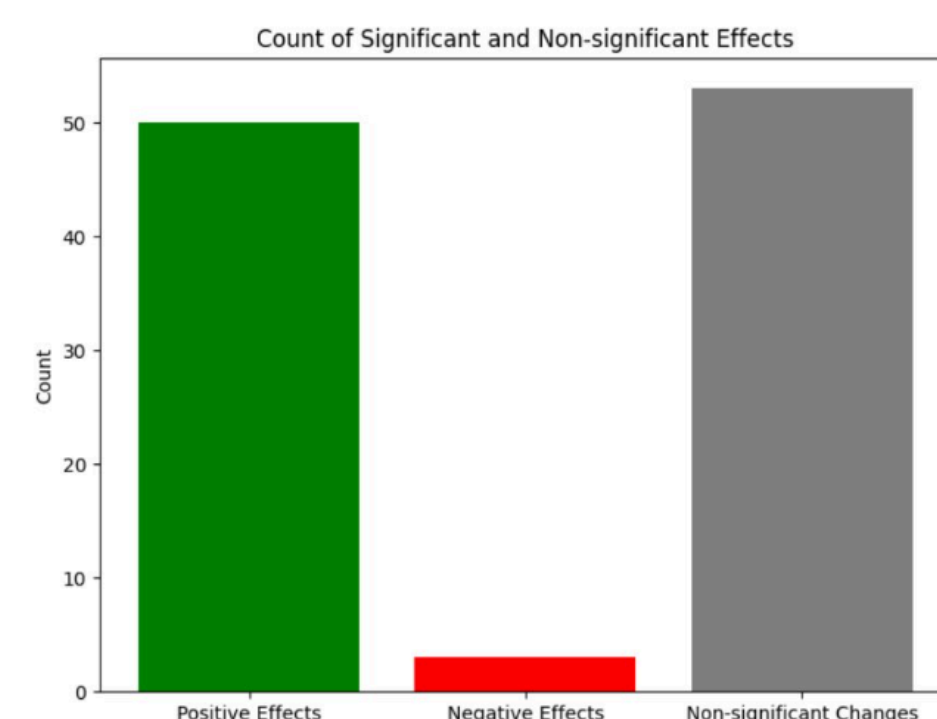
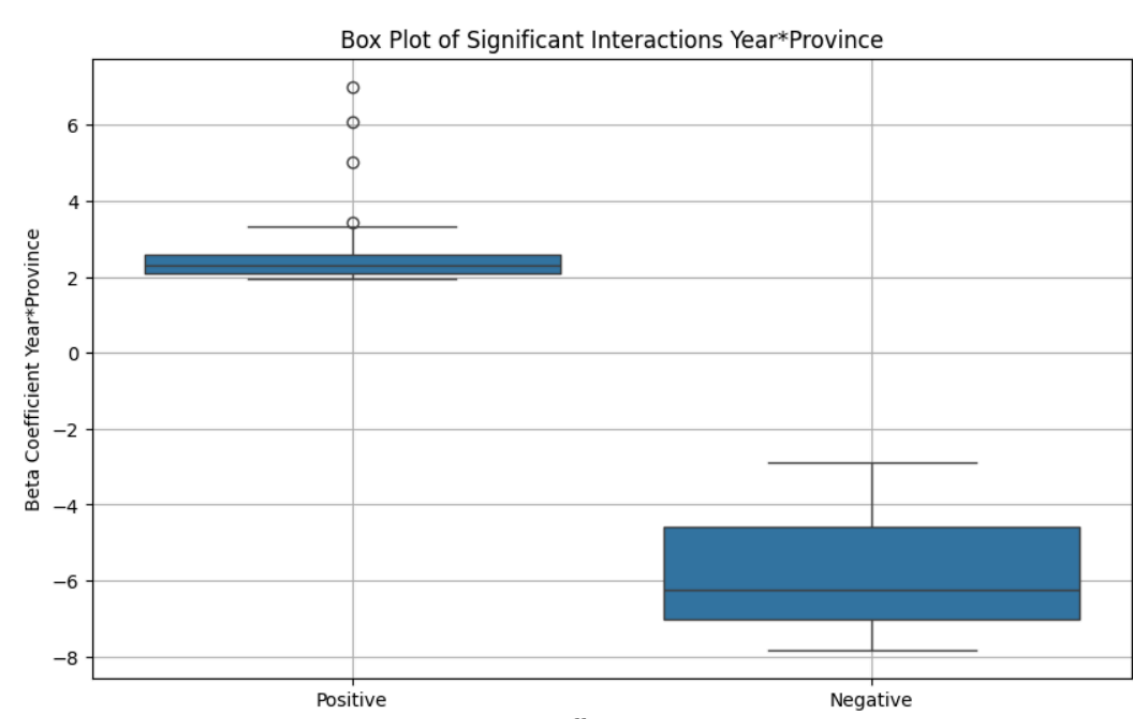
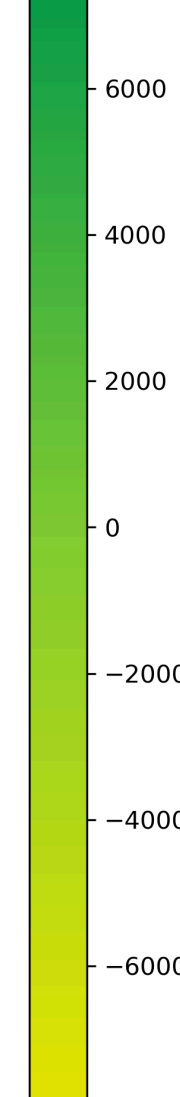
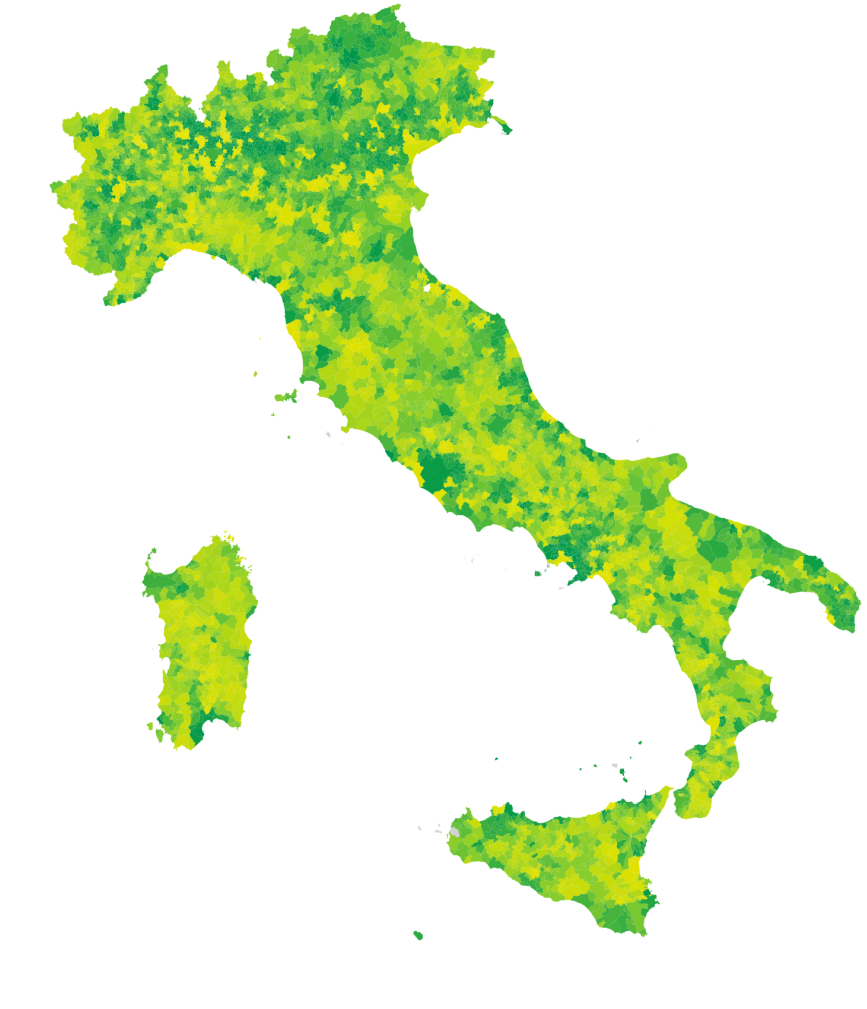
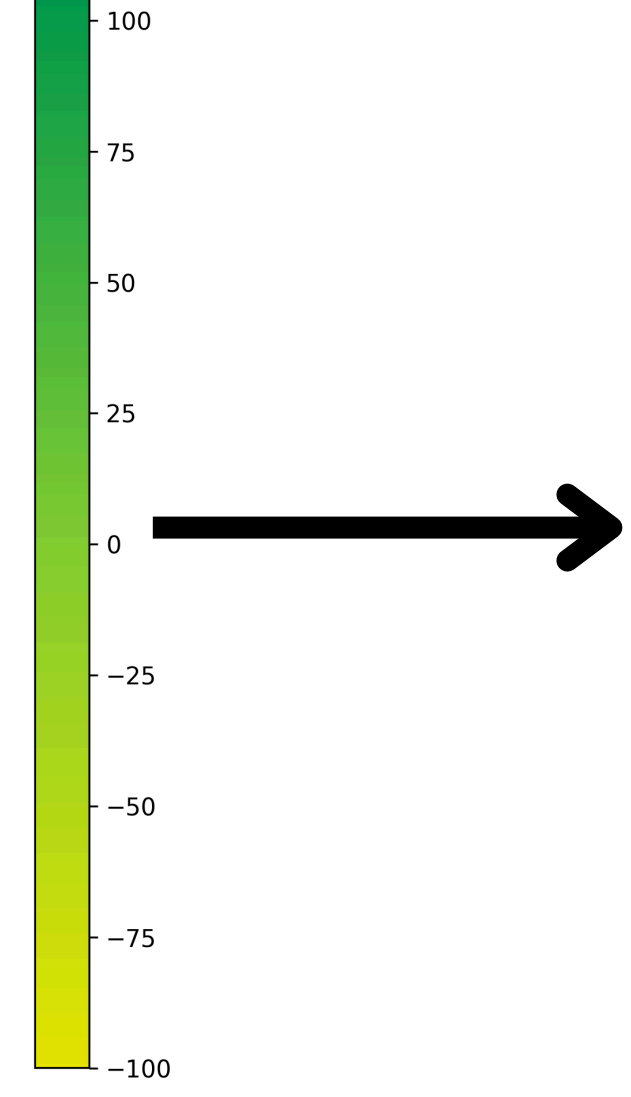
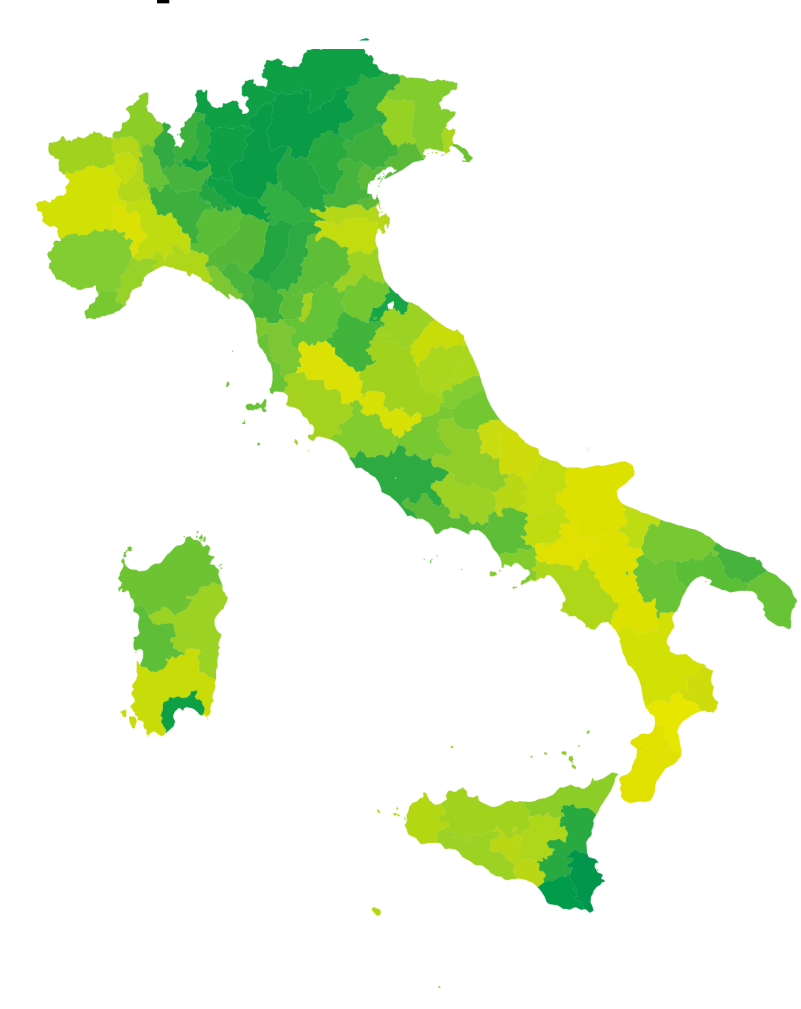
$$S(i, k) = \sum_{l \in L} \begin{cases} +1, & a_{il} > a_{kl} \\ 0, & a_{il} = a_{kl} \\ -1, & a_{il} < a_{kl} \end{cases}$$

$$P_i = w \cdot x_i - (1 - w) \cdot \frac{1}{|N(i)|} \sum_{j \in N(i)} x_j$$

Copeland 2019



Copeland 2022



REFERENCES

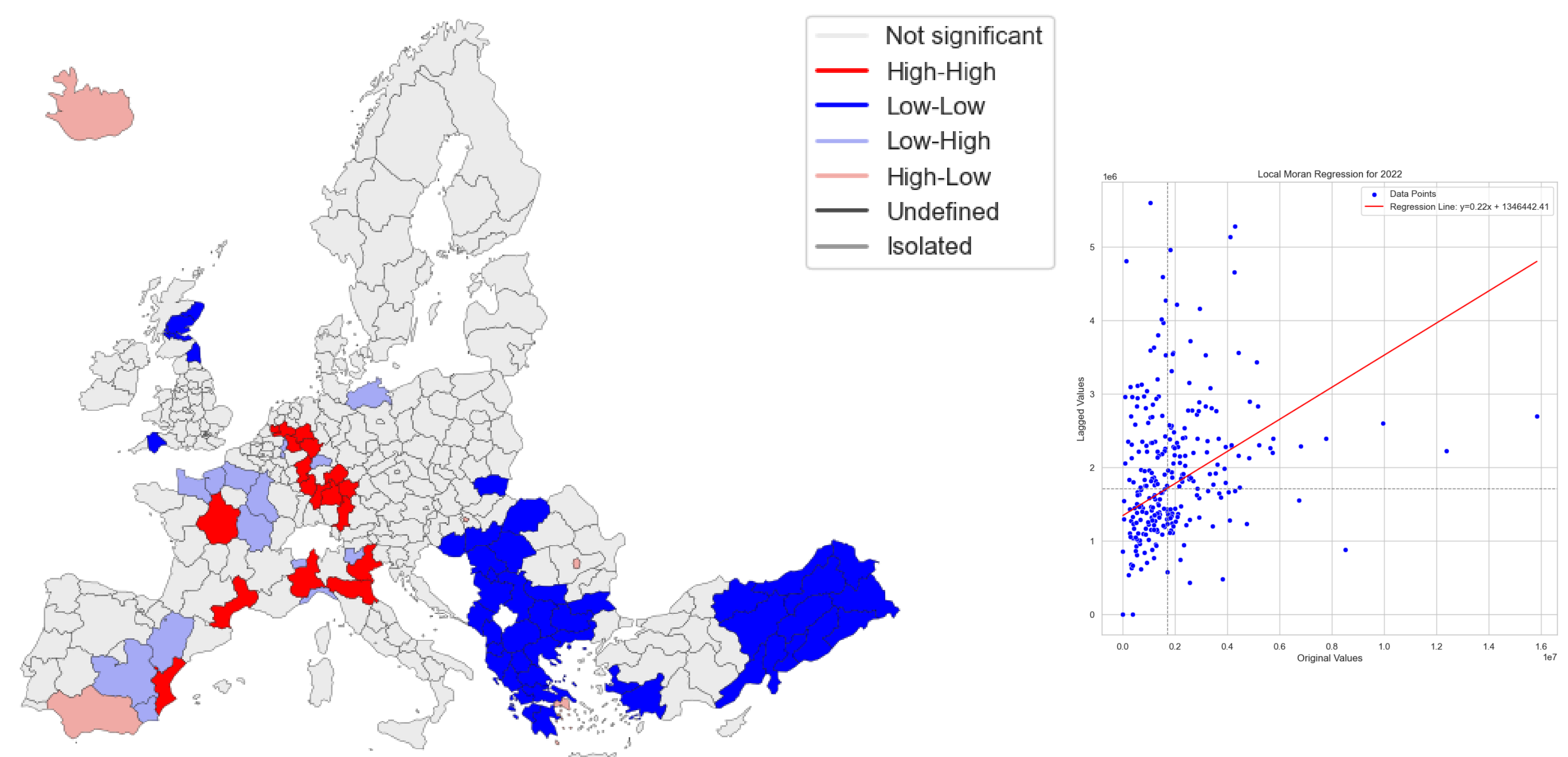
1. Anselin, L. (1955). Local Indicators of Spatial Association—LISA. *Geographical Analysis*, 27(2), 93-115
2. Saari, D. G., & Merlin, V. R. (1996). The Copeland method: I. Relationships and the dictionary. *Economic Theory*, 8(1), 51-76. <https://doi.org/10.1007/BF01212012>
3. Cressie, N., Johannesson, G. (2008). Fixed rank kriging for very large spatial datasets. *Journal of the Royal Statistical Society Series B: Statistical Methodology* 70.1, pp.209-226.

Lisa Maps

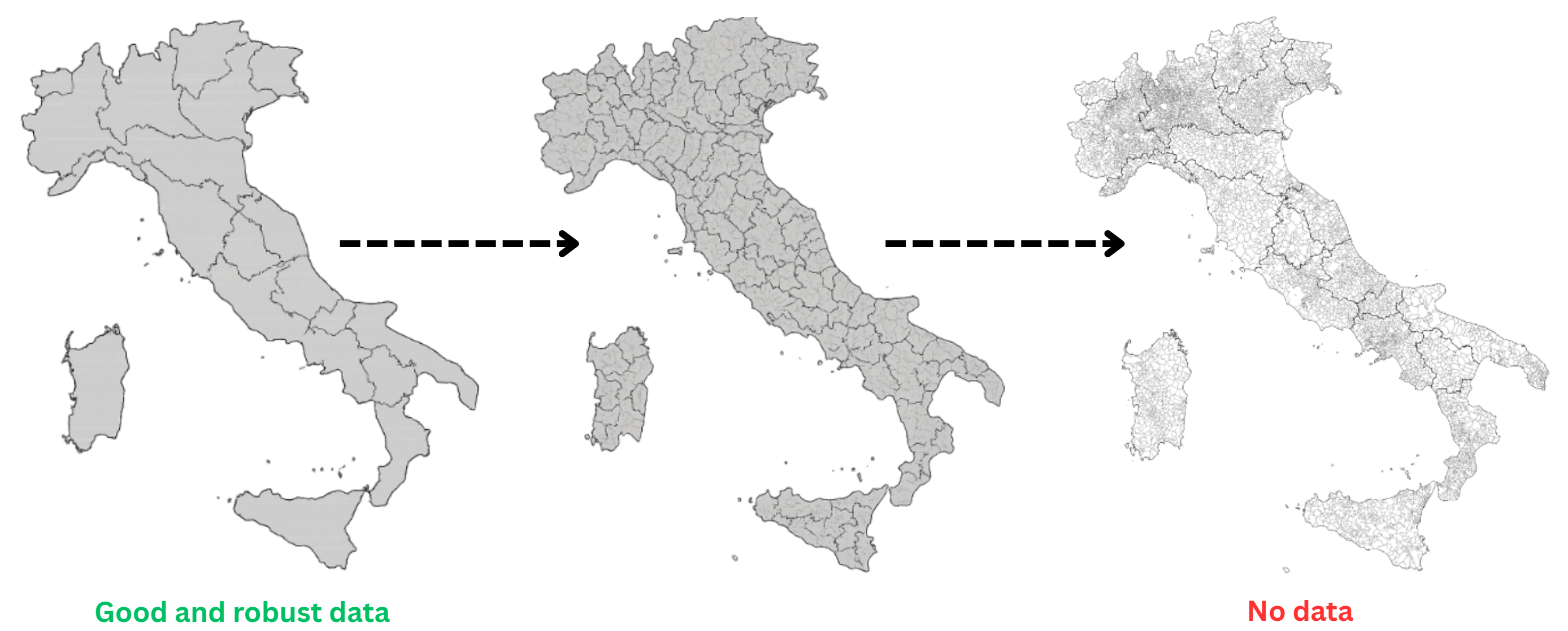
LISA (Local Indicators of Spatial Association) maps help us spot regions that behave similarly to their neighbours, like clusters of high or low growth. This helps us understand local spatial relationships.

$$I = \frac{n}{\sum_{i=1}^n \sum_{j=1}^n w_{ij}} \cdot \frac{\sum_{i=1}^n \sum_{j=1}^n w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Local Moran Cluster Map for 2022

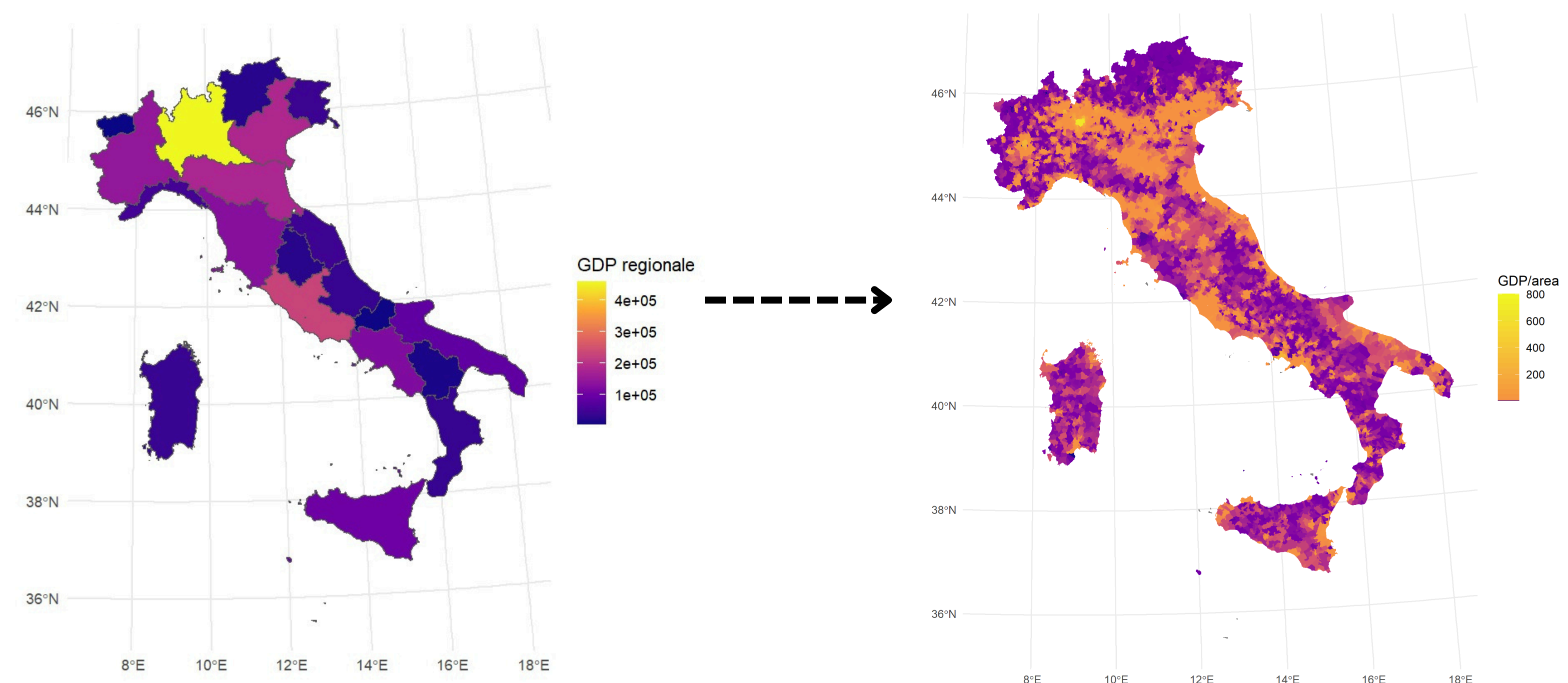


Fixed Rank Kriging



Fixed Rank Kriging provides scalable spatial prediction by combining small-scale basis functions ϕ_j with fine-scale components ξ_s

$$Z_s = \sum_{i=1}^p t_i(s) \alpha_i + \sum_{j=1}^r \phi_j(s) \eta_j + \xi_s$$



Conclusions

1. There exists a spatial influence as can be seen in the Lisa maps.
2. We measured the resilience by using the Copeland methods and to compute the missing GDP data we used the Fixed Rank Kriging.
3. We found the most resilient provinces are in the Lombardy region.